

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal Sentinel-1 SLC IW mode data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5
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S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan

Bridge No. – 899/1

Bridge Chainage – 1027+740

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – MNB

Sar Parameters

Sensor – Sentinel-1 (C-Band)

Beam Mode – IW (Interferometric Wide Swath)

Path – 34

Frame – 519

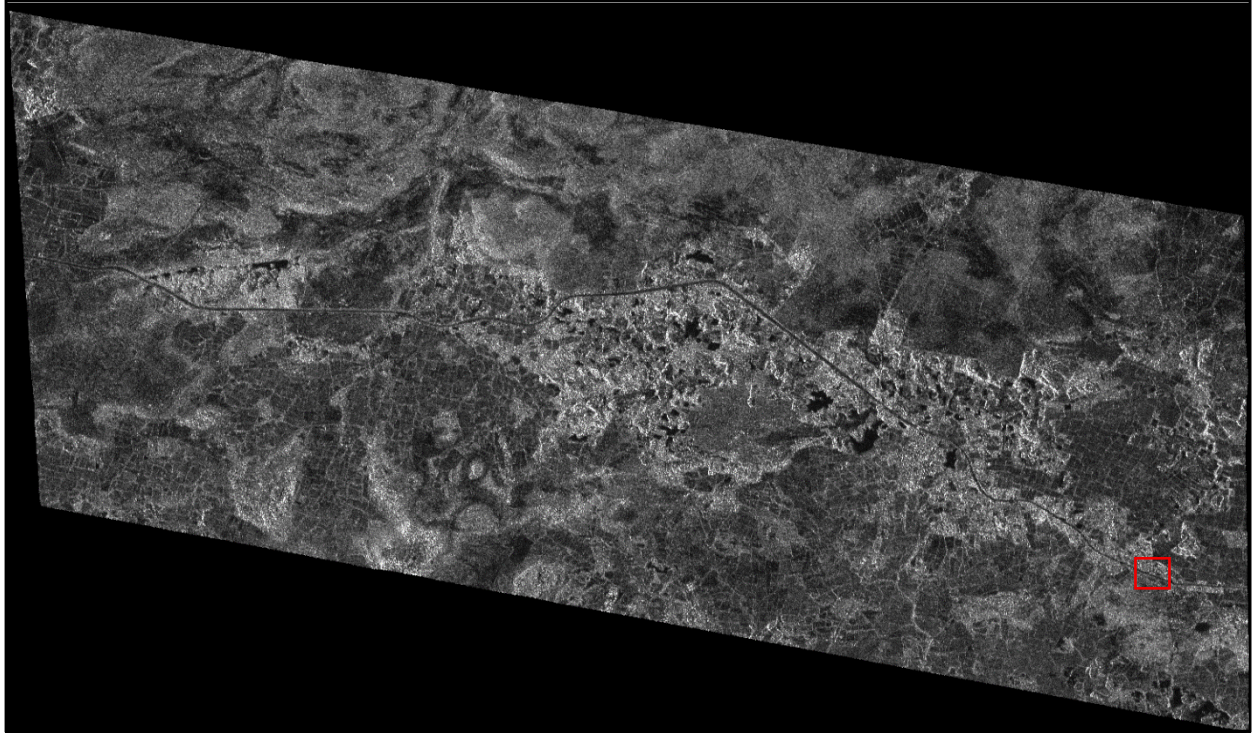
Flight Direction - DESCENDING

Polarization – VV+VH

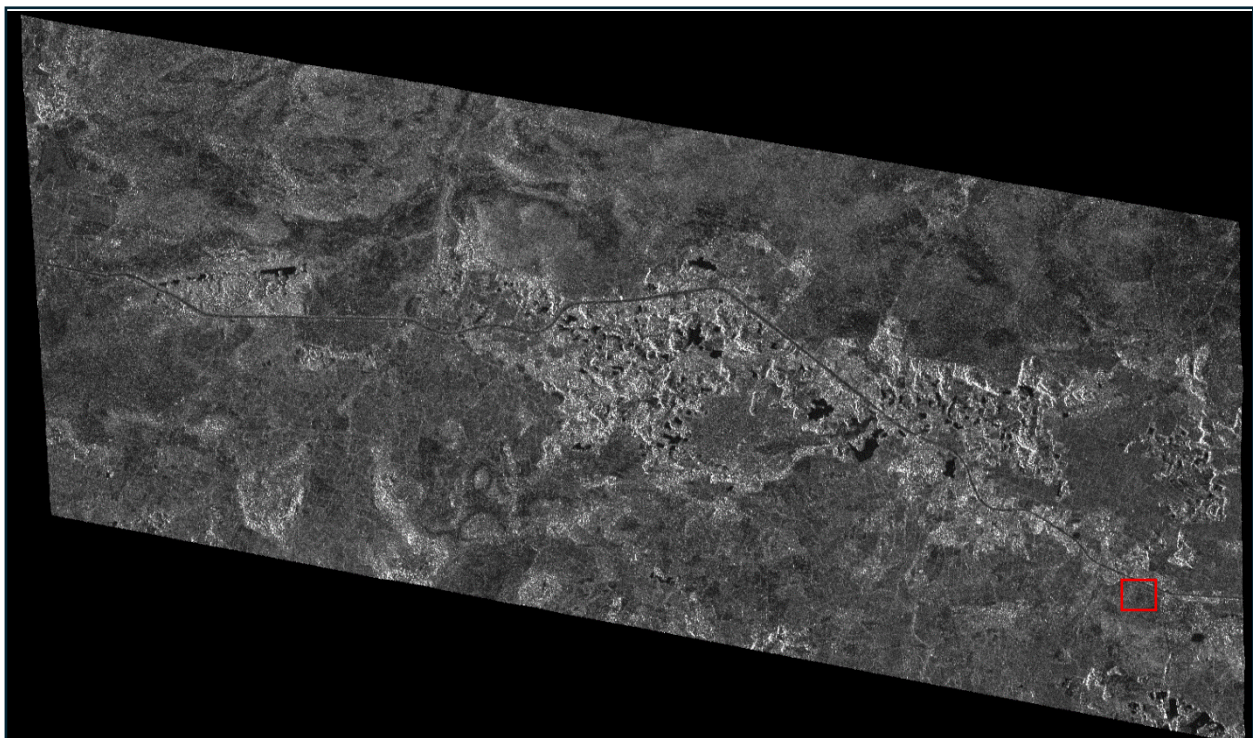
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

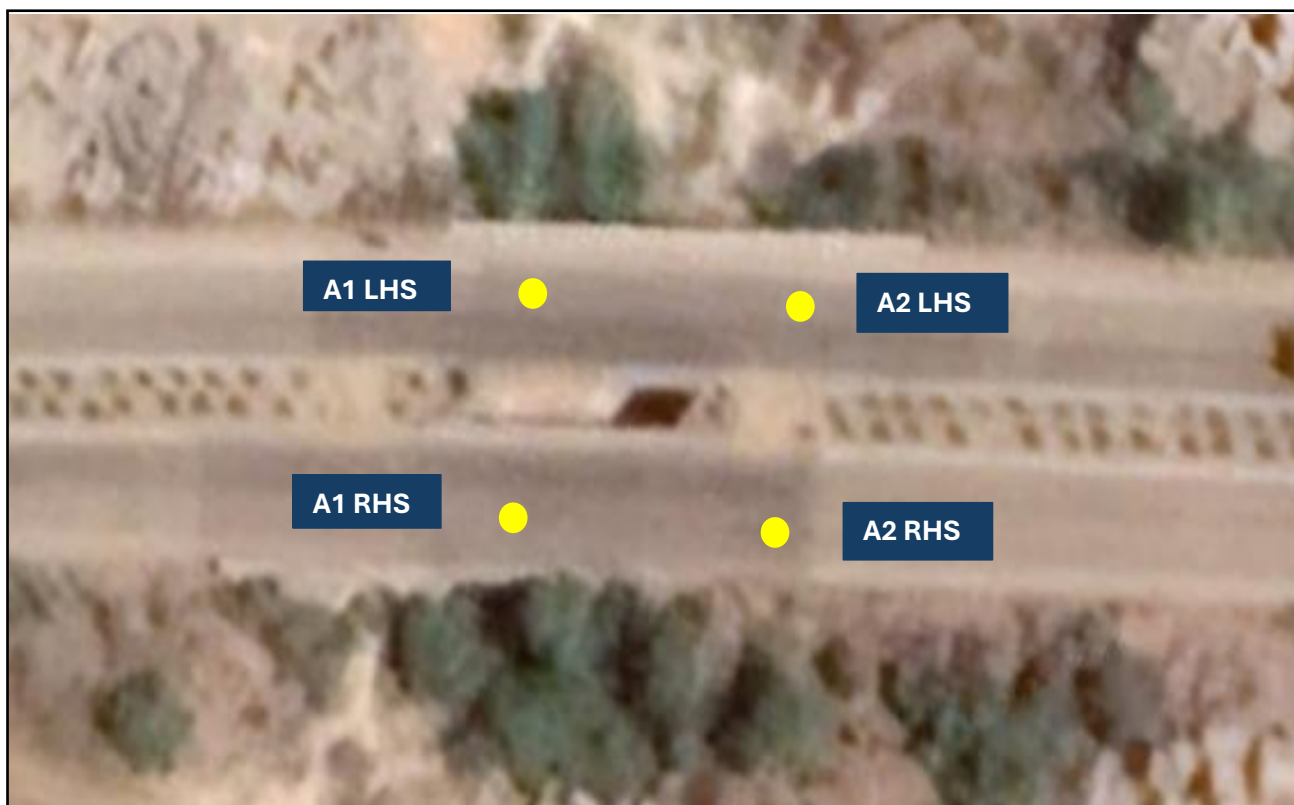


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

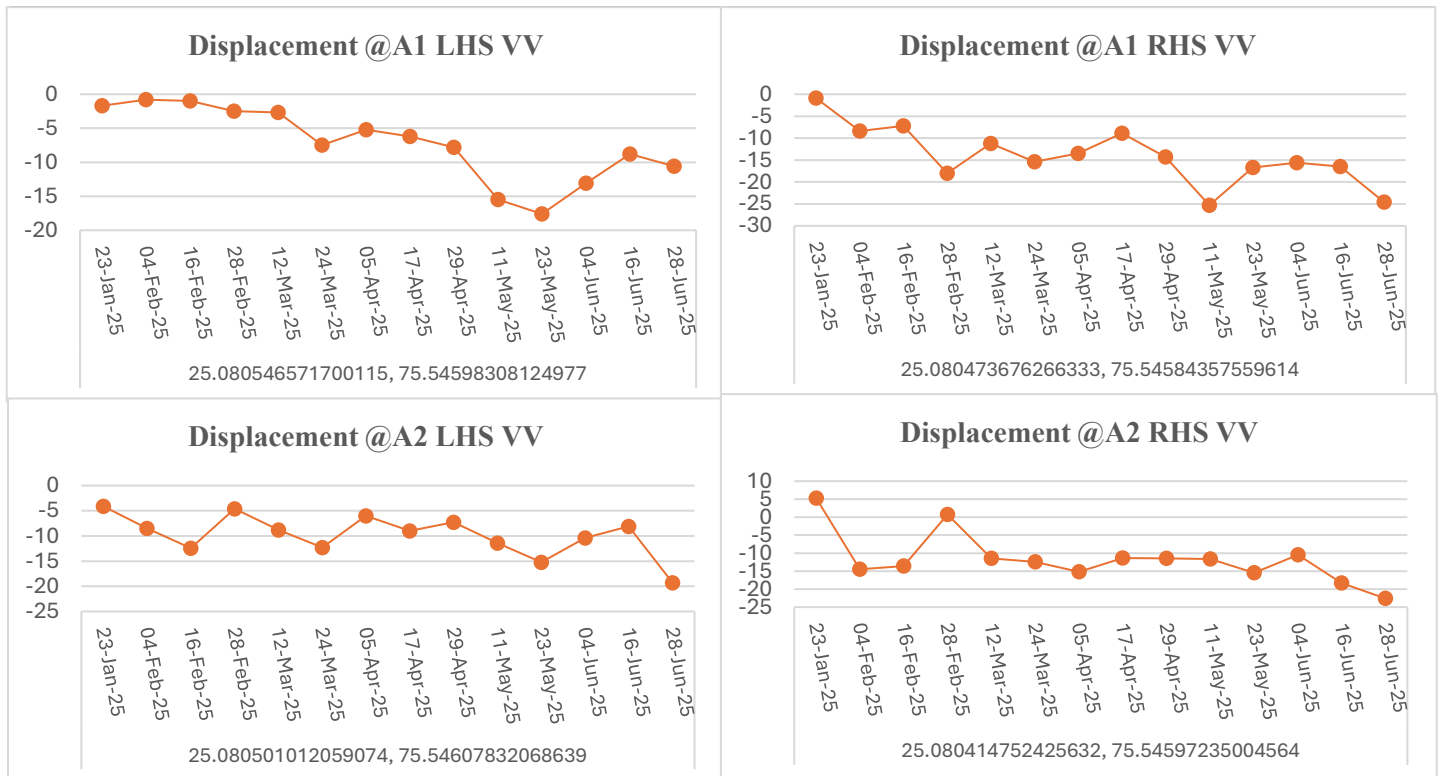
MNB – CH-1027+740

Length – 38.7M

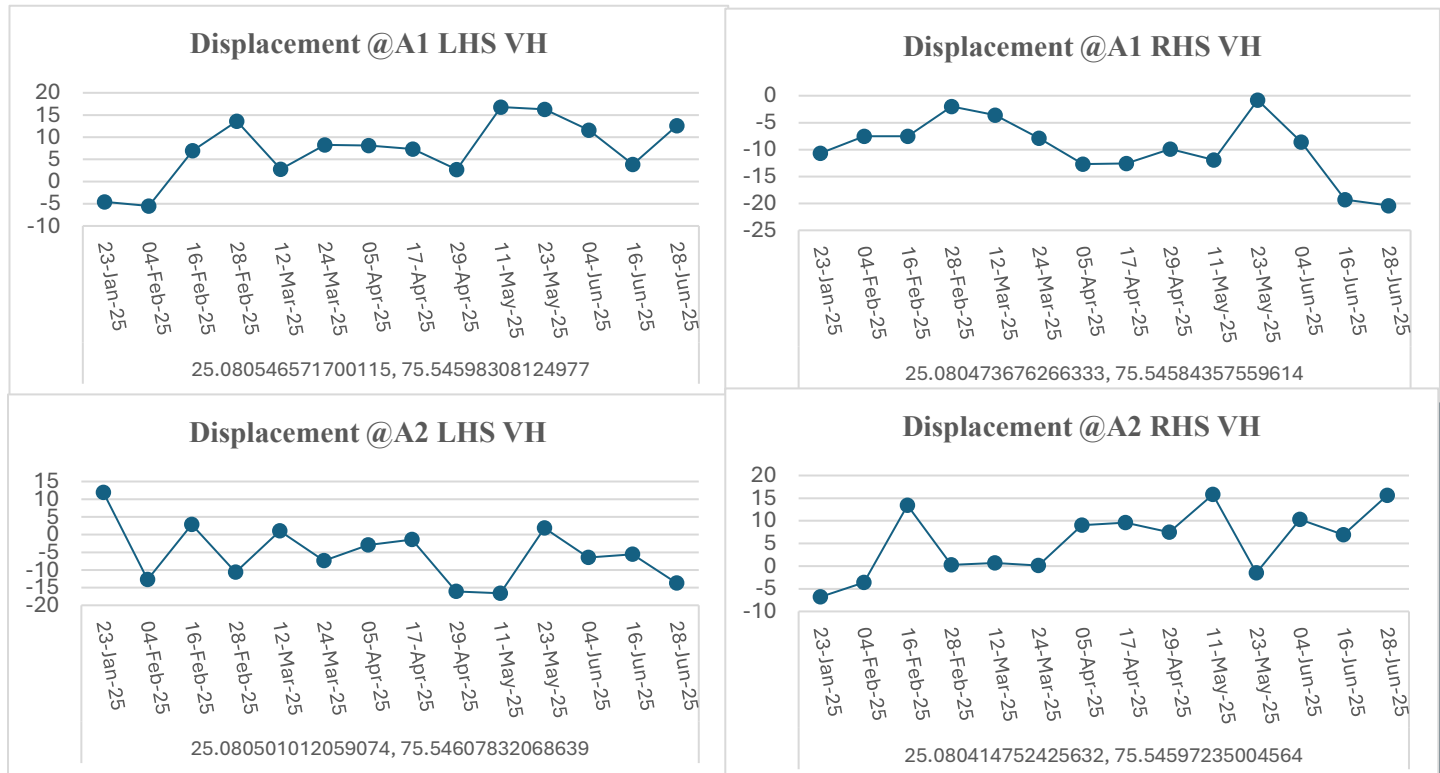


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1	Abutment 1		
A2	Abutment 2		
LHS	Left Hand Side		
RHS	Right Hand Side		

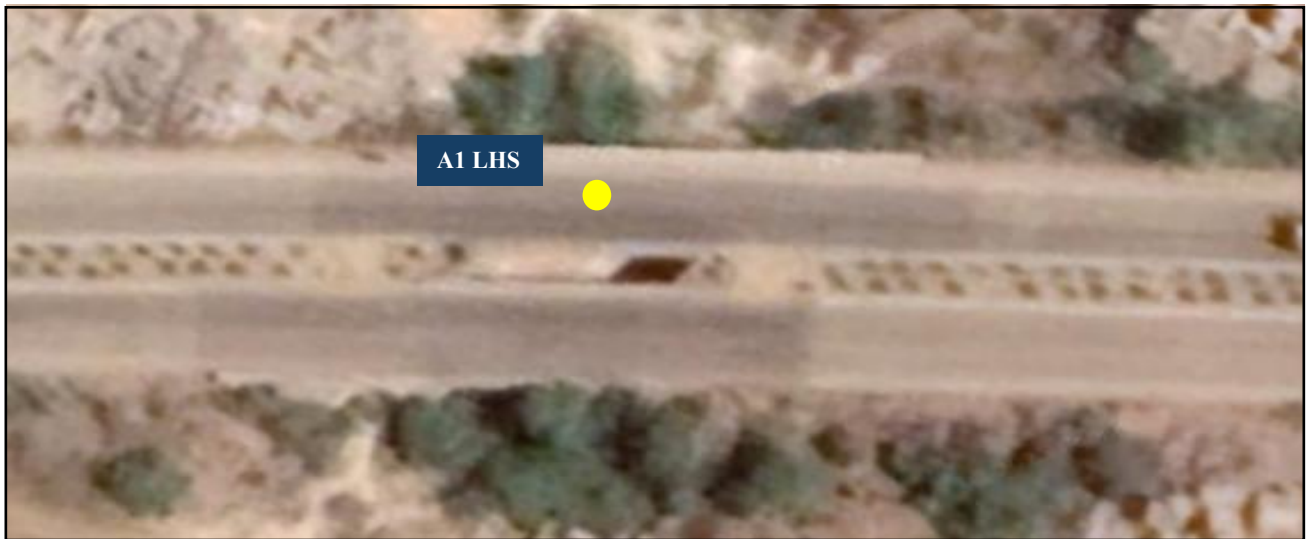
3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



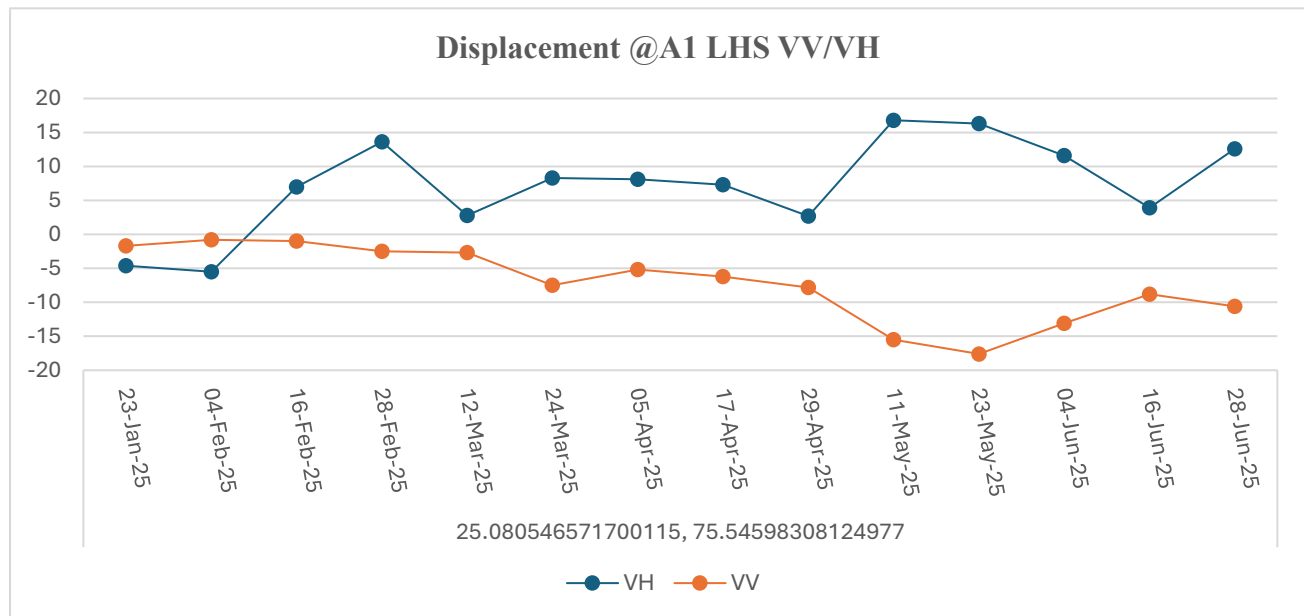
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (25.080546571700115, 75.54598308124977)



Key Observations:

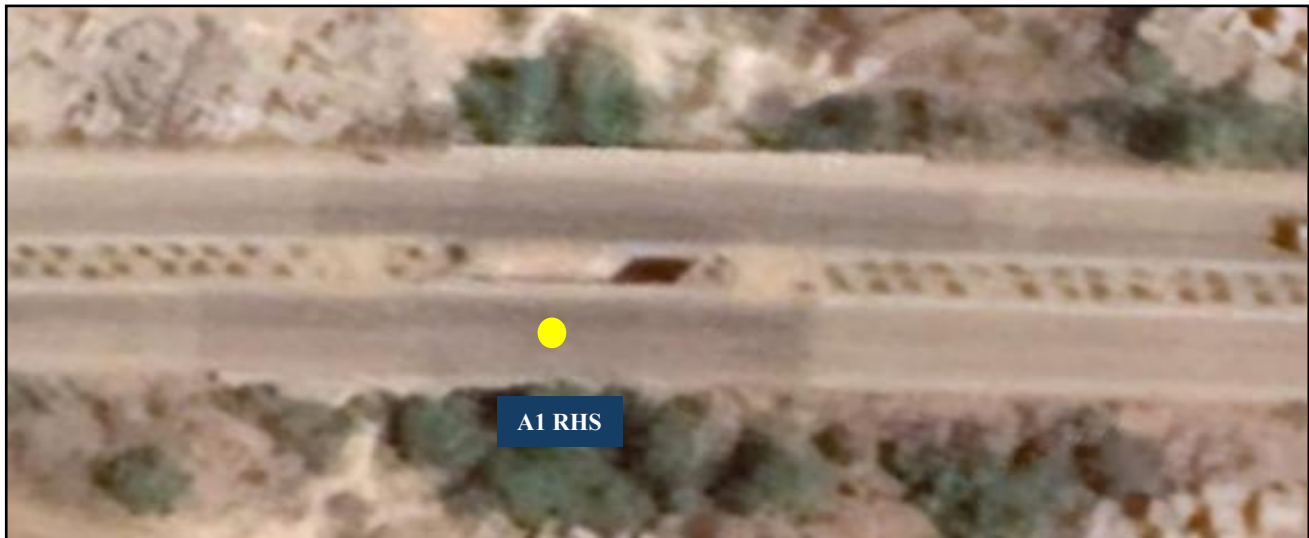
Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement shows a consistently negative trend, indicating ongoing settlement. VV values range approximately between -2 mm and -17 mm.
- **Horizontal Displacement (VH)** Horizontal displacement at A1 LHS is predominantly positive throughout the monitoring period, indicating sustained lateral movement in one direction.

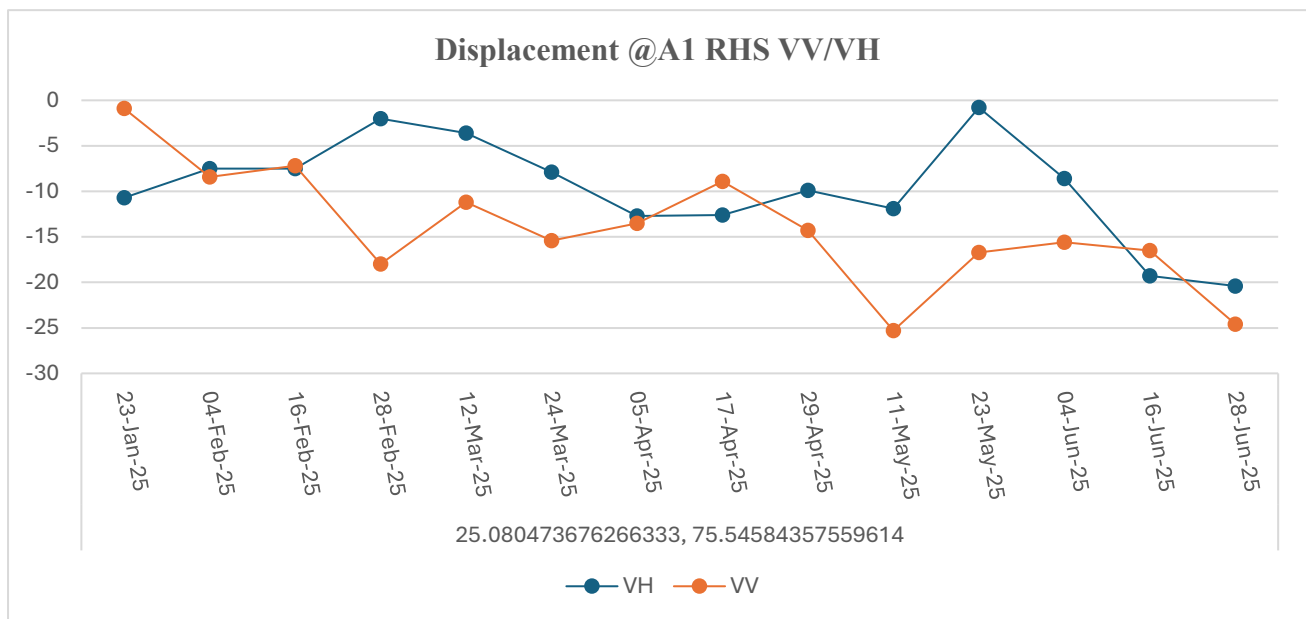
Critical Events & Anomalies:

- **February 2025:** VH reaches an early local peak of approximately $+14$ mm, while VV remains close to neutral (-2 mm), suggesting dominant lateral movement with minimal settlement at this stage.
- **May 2025:** VV reaches its maximum negative displacement (-17 mm), representing the most critical settlement phase at this location.
- **June 2025:** VV shows partial recovery toward less negative values (-8 to -10 mm), while VH decreases briefly before increasing again toward $+12$ mm, suggesting stabilisation of settlement with continued lateral response.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (25.080473676266333, 75.54584357559614)



Key Observations:

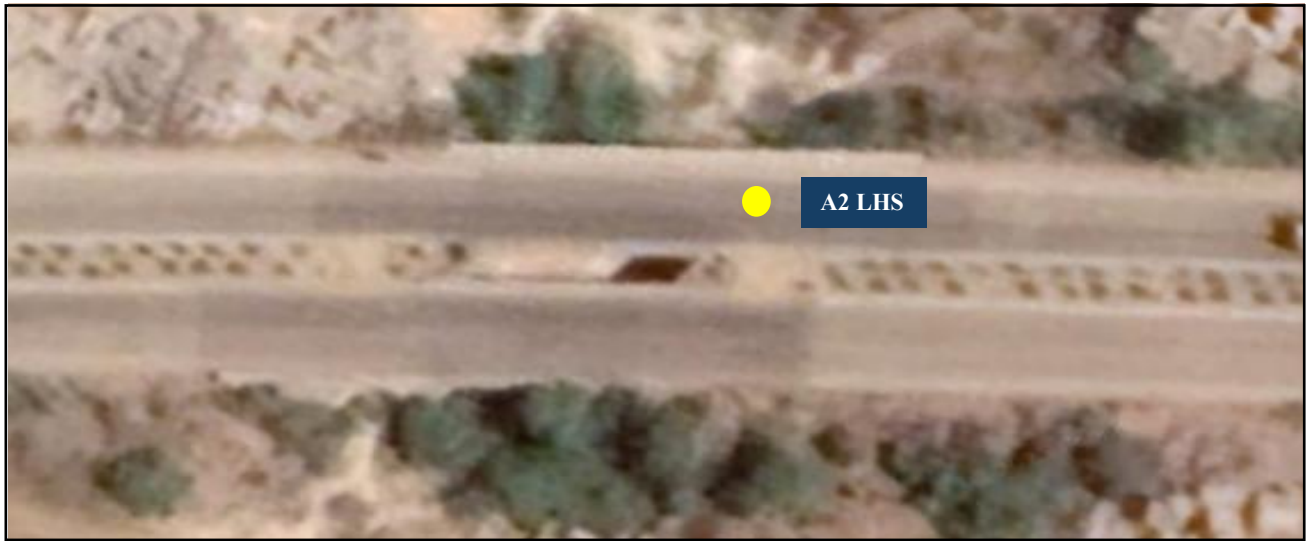
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement remains consistently negative throughout the monitoring period, indicating sustained lateral movement in a single dominant direction.
- **Vertical displacement (VV)** Vertical displacement is also consistently negative, indicating continuous settlement behaviour rather than uplift. VV ranges approximately from –5 mm to –27 mm, with larger variability compared to VH and several pronounced settlement phases

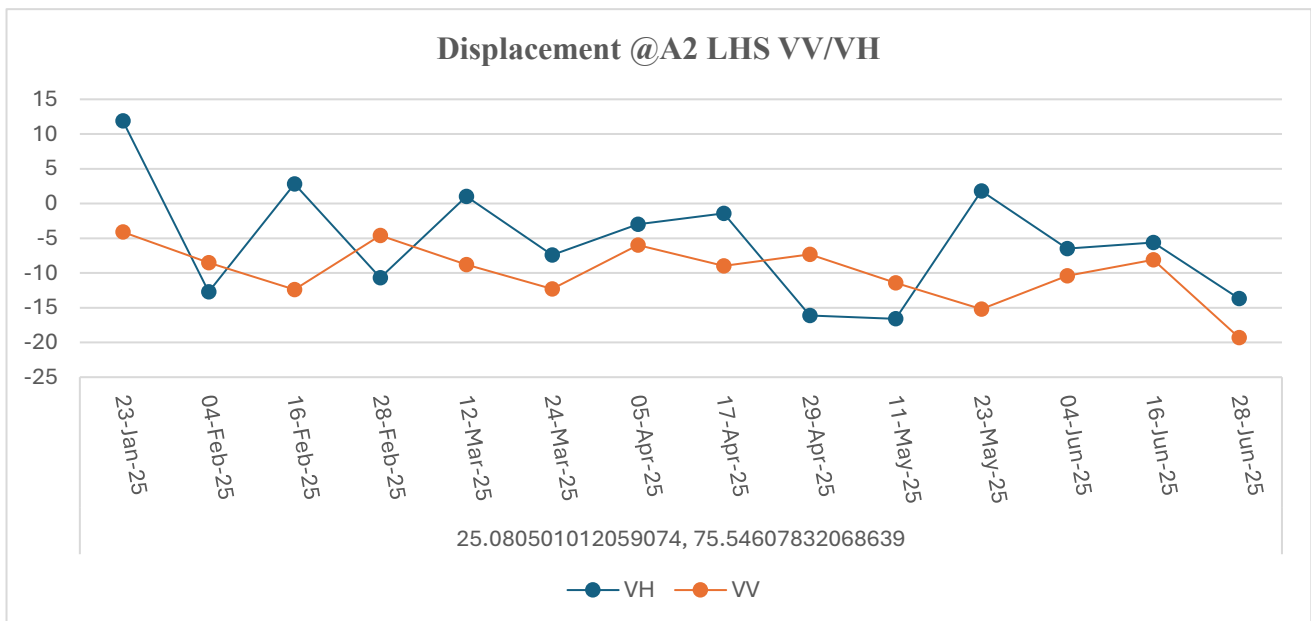
Critical Events & Anomalies:

- **February 2025:** VH shows a temporary reduction in magnitude (around –3 to –4 mm), while VV exhibits a sharp increase in settlement. This divergence suggests vertical movement dominating while lateral movement temporarily stabilizes.
- **June 2025:** Both VH and VV show renewed downward trends, with VH reaching –18 to –20 mm and VV remaining high at –22 to –27 mm.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (25.080501012059074, 75.54607832068639)



Key Observations:

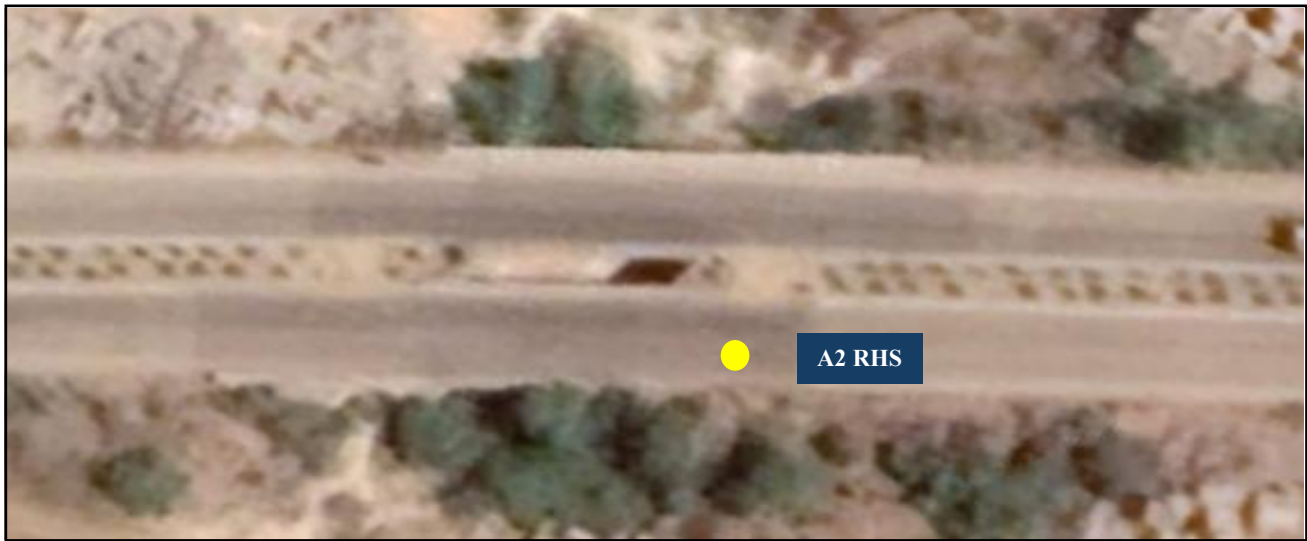
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement shows bidirectional behaviour, fluctuating between positive and negative values over the monitoring period.
- **Vertical Displacement (VV)** Vertical displacement remains consistently negative throughout the monitoring period, indicating continuous settlement behaviour with no uplift reversal.

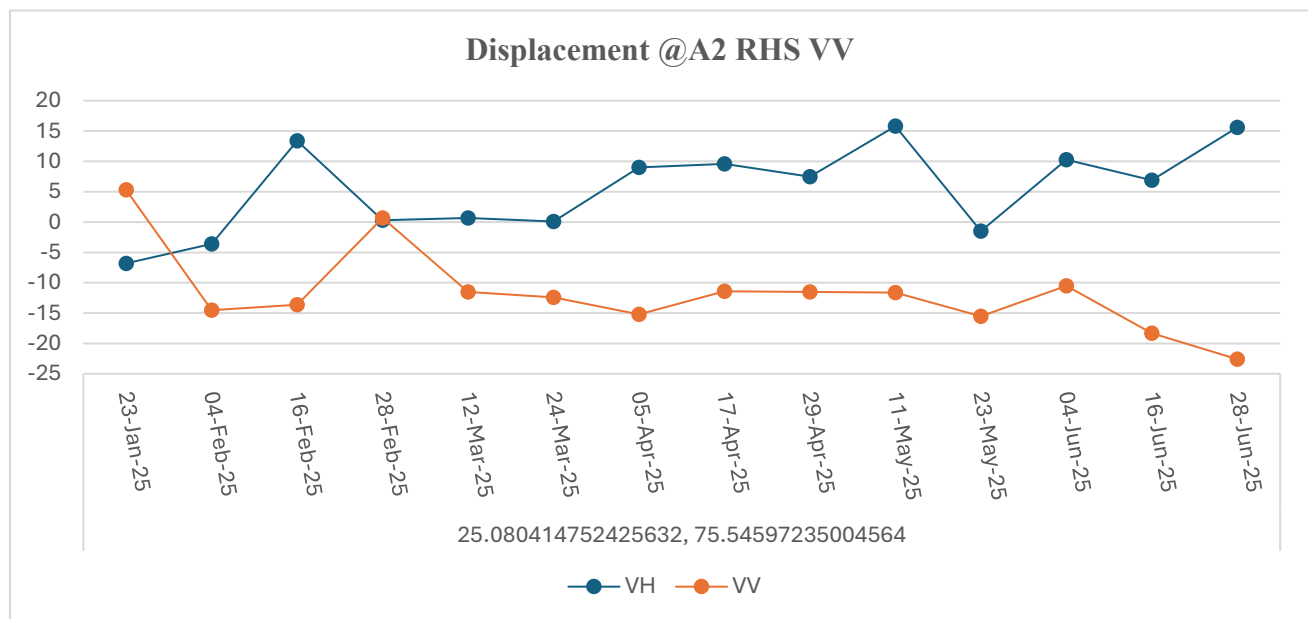
Critical Events & Anomalies:

- **February 2025:** VH drops sharply to approximately -13 mm, coinciding with increased settlement in VV (around -10 to -12 mm). This indicates a transition phase, where lateral movement reverses direction and vertical settlement accelerates.
- **June 2025:** Both VH and VV trend downward, with VH settling near -14 to -15 mm and VV reaching -18 to -20 mm. This reflects combined lateral instability and progressive vertical settlement, indicating ongoing deformation without recovery.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (25.080414752425632, 75.54597235004564)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** exhibits a clear negative trend over time, indicating progressive settlement. VV values range approximately from +9 mm to –20 mm.
- **Horizontal (VH) displacement** shows predominantly positive behaviour, indicating lateral movement largely in a single dominant direction. VH values range approximately from –5 mm to +15 mm, with moderate oscillations and several short-term reversals into slight negative movement

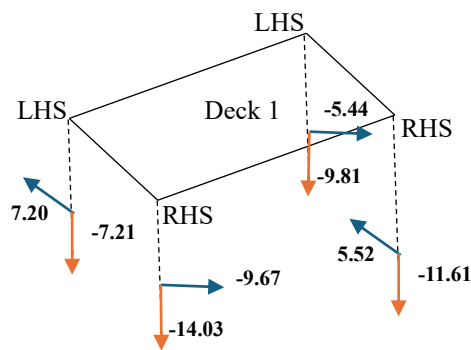
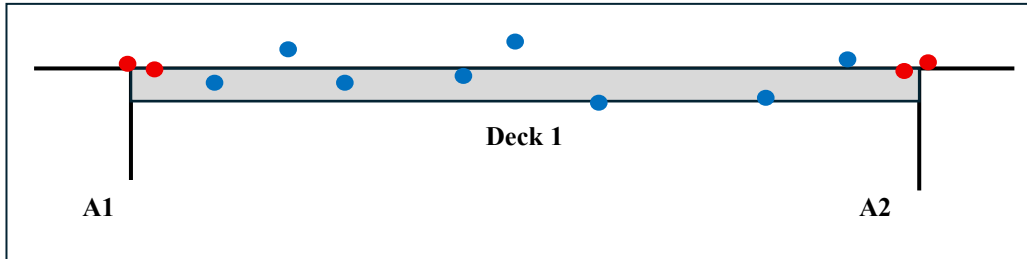
Critical Events & Anomalies:

- **February 2025:** VH stabilizes near 0 mm, while VV briefly reduces in settlement magnitude (near –5 mm). This reflects a temporary vertical response moderation with lateral movement remaining subdued.
- **June 2025:** VH recovers to moderate positive values (+8 to +15 mm), while VV reaches its maximum settlement, approximately –22 to –23 mm. This reflects sustained vertical settlement with resumed lateral movement, indicating ongoing deformation without stabilization.

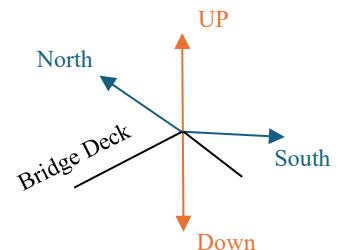
ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Bridge Deck Slab Analysis



Vertical Displacement Behaviour:

- The bridge deck slab exhibits pronounced non-uniform vertical displacement across both the Left-Hand Side (LHS) and Right-Hand Side (RHS), indicating differential vertical response of the deck slab. vertical displacement pattern suggests uneven load transfer and stiffness variation, leading to differential vertical movement across the deck width.

Horizontal Displacement Behaviour:

- Horizontal displacement demonstrates clear directional variability, reflecting torsional effects and asymmetric structural behaviour of the bridge deck slab. The maximum horizontal displacement is observed near the central deck region, reaching approximately +9.67 mm, indicating significant lateral movement at mid-span

CONCLUSION:

- The bridge deck slab displays a highly complex deformation response, characterized by simultaneous vertical settlement and lateral torsional movement. Overall, the deformation pattern warrants continued monitoring and correlation with loading stages, temperature effects, and long-term structural behaviour to assess serviceability and durability performance.